FORMULA SHEET

Mean:

$$M = \frac{\sum_i^n x_i}{n}$$

Weighted Mean:

$$M_w = \frac{\sum_i^n w_i x_i}{\sum_i^n w_i}$$

Geometric Mean:

$$G = \sqrt[n]{\prod_{i=1}^{n} x_i}$$

Quadratic Mean:

$$Q = \sqrt[2]{\frac{\sum_{i=1}^n x_i^2}{n}}$$

Percentile:

$$L_y = (n+1)\frac{y}{100}$$

MAD:

$$MAD = \frac{\sum_{i=1}^n |x_i - M|}{n}$$

Standard Deviation:

$$\sigma(X) = \sqrt{\frac{\sum_{i=1}^n (x_i - M)^2}{n}}$$

Variance:

$$\sigma^2(X) = \frac{\sum_{i=1}^n (x_i - M)^2}{n}$$

Expected Value, Variance for discrete distrib:

$$E(X) = \sum_{i=1}^{n} x_i f(x_i)$$

Expected Value, Variance for continuous distrib:

$$E(X) = \mu_x = \int_{-\infty}^{+\infty} x \cdot f(x)dx$$

$$V(X) = \int_{-\infty}^{+\infty} (x - \mu_x)^2 \cdot f(x)dx$$

Normal distribution:

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

Z-Score:

$$Z = \frac{X - \mu}{\sigma}$$

Covariance:

$$cov(x, y) = \sigma_{XY} = \frac{\sum_{i=1}^n (y_i - \mu_Y)(x_i - \mu_X)}{n}$$

Correlation:

$$-1 \leq \underbrace{\frac{\sigma_{XY}}{\sigma_X \sigma_Y}}_{\rho_{XY}} \leq 1$$

Simple Linear Regression

$$\hat{\beta}_0 = \mu_Y - \hat{\beta}_1 \mu_X \quad ; \quad \hat{\beta}_1 = \frac{\sigma_{XY}}{\sigma_X^2}$$

Expected Return and variance of a portfolio:

$$R_p = \sum_{i=1:N} w_i R_i$$

$$Var(R_p) = \sum_{i=1:N} \sum_{j=1:N} w_i w_j Cov(R_i, R_j)$$

Beta $\boldsymbol{\beta_{i,m}}$:

$$\beta_{i,m} = \frac{COV(R_i, R_m)}{VAR(R_m)}$$

CAL:

$$E(R) = R_f + \frac{E(R) - R_f}{\sigma_p}\sigma$$

Utility:

$$U = E(R) - 0.5 \times A \times Var(R)$$

Capital Allocation Weight:

$$y^* = \frac{E(R_P) - R_F}{A \times Var(R_P)}$$

Security Market Line (SML):

Investment Management

$$E(R_i) = R_F + \beta_{i,M}[E(R_M) - R_F]$$

Market Model Regression:

$$R_{i,t} = a_i + b_i\left(R_{M,t} - R_{F,t}\right) + \varepsilon_{i,t}$$

Characteristic Line Regression:

$$R_{i,t} - R_{F,t} = a_i + b_i\left(R_{M,t} - R_{F,t}\right) + \varepsilon_{i,t}$$

Expected Return (CAPM):

$$E(R_i) = R_F + \beta_{i,M}\lambda_M$$

Sharpe Ratio:

$$S_P = \frac{E(R) - R_f}{\sigma_p}$$

Jensen's alpha:

$$\alpha_P = E(R_P) - R_F - \beta_{P,M}[E(R_M) - R_F]$$

Appraisal Ratio:

$$A_P = \frac{\alpha_P}{\sigma_{\varepsilon_P}}$$

Investment Management